



OPEN Mixture of checkpoint experts for explainable seizure detection using wearable devices

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The current gold standard for detecting epileptic seizures is in-hospital video-Electroencephalography (vEEG), but vEEG is resource-intensive and imposes considerable burdens on patients and caregivers. Wearable devices offer an alternative to monitor seizures over long periods in a home environment and Machine Learning (ML) models can be used to analyze the data and identify seizure activity. The best performing ML models tend to be black-box models which do not provide a rationale for their decision, making it difficult to audit their mistakes and limit their real-world utility. To combat this, we propose a novel ML algorithm, Mixture of Checkpoint Experts (MoCE). MoCE provides direct insights into the behavior of a model at both the global (overall performance) and local (individual predictions) level that are not available with classic black-box models. Through a study conducted in 14 patients with epilepsy using wrist-worn devices, we demonstrate how this transparency is beneficial to both clinicians and data scientists, who can audit individual predictions for the reason of a decision, providing greater information about the model and its behaviors. Our experiments reveal that MoCE is capable of detecting seizures better than existing neural networks by retaining equivalent performance in recall with statistically significant improvement in false alarm rate while simultaneously addressing the issues of black-box models that many clinicians face.

Keywords Explainable machine learning, Mixture of experts, Seizure detection, Wearable devices

Epilepsy is a common neurological disorder characterized by recurrent seizures that can manifest as temporary disruptions in muscle movements, behaviors, sensations, or levels of awareness¹. Affecting over 70 million people worldwide, Epilepsy remains a significant contributor to morbidity and mortality, despite advancements in treatment strategies². An accurate account of how frequently a patient experiences seizures is essential to the successful management of Epilepsy. These records are used to assess seizure burden, estimate recurrence risk, and evaluate therapeutic responses. Video-Electroencephalography (vEEG) is the gold standard for seizure detection and prior work in Deep Learning has looked into detecting seizures through explainable models^{3,4}. However, vEEG is resource-intensive, limited in duration, and imposes a considerable burden on patients and caregivers⁵. For home-based seizure tracking, seizure diaries provide a more feasible option, however, these self-report data have been shown to be inaccurate as patients may not recall a seizure or miss those occurring at night, resulting in underreporting^{6,7}.

Seeking an alternative to unreliable seizure diaries, researchers have found potential in leveraging wearable technology and machine learning (ML) models to identify and track seizures in a non-invasive manner⁸. Wearable devices worn on the wrist are capable of capturing multimodal physiological signals, including blood volume pulse (BVP), heart rate (HR), heart rate variability (HRV), accelerometry (ACC), skin temperature, and electrodermal activity (EDA), all of which have demonstrated correlations with seizure activity⁹. The application of machine learning models to these data has demonstrated promising performance in detecting and even forecasting Generalized Tonic-Clonic (GTC) seizures—a seizure characterized by the stiffening of the body, followed by rhythmic jerking^{10,11}.

Typically, researchers rely upon black-box model architectures where the rationale for a decision is obscured. Traditional ML algorithms such as Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) are prominent in seizure detection using EEG recordings^{12–14}. However, in the realm of wearable devices, more complex neural network architectures such as Convolutional Neural Networks (CNN) and

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Recurrent Neural Networks tend to excel due to their advanced ability to detect patterns in spatial and sequential data^{15–19}. Despite these advancements, however, there remains a need for greater transparency surrounding many of these models²⁰.

To combat these black-box models, eXplainable Artificial Intelligence (XAI) techniques have been increasingly applied in domains such as healthcare, where insights into the models' decision making process are critical²¹. XAI efforts are usually divided into two groups: interpretability and explainability. Interpretability focuses on designing models capable of conveying precisely how a decision is made. In this field, transparency is of the utmost importance where no approximations are necessary and the model's training process can be quickly and easily deciphered. Explainability, on the other hand, focuses on extracting a model's decision-making process. This is often accomplished by fitting post-hoc models to approximate the relationship between inputs and outputs when the initial model is too complex for humans to understand. SHAP values are a common example of post-hoc explainability²².

In both cases, the result is an explanation that describes how individual features impact the final prediction. However, there is no one best approach. The requirements of XAI can be ambiguous and heavily dependent upon the specific context²³. Furthermore, different stakeholders, such as data scientists versus clinicians, will have unique model explanation needs, that the narrow scope of these techniques may be unable to address²⁴. Black-box models with post hoc explainers are popular because of their superior predictive performance. These models specialize in identifying "hidden patterns" within the data that more straightforward, interpretable models are incapable of finding²⁵. On the other hand, post-hoc explainability cannot fully approximate a black-box model, which often leads to biased and unfair conclusions²⁶. As a result, these models are usually inappropriate for high-risk domains where proper auditing of mistakes is vital.

In this paper, we propose a novel machine learning technique, **Mixture of Checkpoint Experts (MoCE)**, which retains the predictive power of black-box models while introducing partial transparency into their decision-making process. We propose a method which can be applied to any iteratively trained black-box model. Our proposal improves upon existing explanation techniques by offering direct insights into a model's inner-workings, not approximations which may be biased or misleading, while retaining predictive performance. Through a robust case study, we answer the following research questions:

- RQ1 Does **MoCE** improve the predictive performance of a standard black-box neural network?
- RQ2 Can **MoCE** be used to understand a neural network's global decision-making patterns?
- RQ3 Is **MoCE** able explain the rationale for individual predictions in a way that is beneficial to both data scientists and clinicians?

Background and significance

In existing literature, the issue of seizure detection has been addressed through various neural network architectures including Long-Short Term Memory (LSTM) models and Convolutional Neural Networks (CNN)^{27,28}. Although neural networks have shown potential for identifying GTC seizures, their black-box nature complicates their usage in a clinical setting²⁹. Transparent models allow a human domain expert to audit a model's output and refine its decision-making process, making them more appropriate for high-risk domains such as the clinical setting³⁰. However, traditional transparent models suffer from lower predictive power, especially when dealing with complex data types such as multivariate time series.

Additionally, in a typical neural network, a model is fit over a number of epochs by adjusting the weights applied to the features at each step with the goal of minimizing a global loss function. During this training process, the model might learn and unlearn many patterns that perform poorly at a global level. However, we contend that these patterns may provide valuable insights for niche cases and are better suited for a small subset of the data. Checkpoint ensembles counteract this by building an ensemble learner from various epochs or checkpoints during the training process^{31,32}. These ensembles have previously been used in the clinical setting to classify gastrointestinal diseases from endoscopic images³³. Although checkpoint ensembles were shown to have improved performance over neural networks, they continue to suffer from limited transparency.

Mixture of Experts (MoE) is another ensembling technique in which multiple, unique models are fit and their outputs combined into a final prediction³⁴. Ismail et al.³⁵ proposed a variant: Interpretable MoE, which instead of combining the predictions of multiple models, uses expert assignment to predict a single expert for each sample, providing an interpretable ensemble. Prior work has combined Interpretable MoE with individual meta-learners to introduce the concept of partial model transparency, retaining predictive performance of black-box methods while adding interpretability^{36,37}.

Meta-learning is frequently used for error correction with applications in transfer learning, neural networks, and few-shot learning³⁸. Differently, we use meta-learning for error anticipation, attempting to predict model performance using a combination of the original feature space, meta-features, and model predictions. Prior work has detailed using meta-learners for classifier selection^{37,39,40}.

Our proposed method combines the ideas of Checkpoint Ensembles and Interpretable MoE into a single algorithm. **MoCE** improves upon existing XAI techniques by providing direct insights into the model's internal behavior while simultaneously exploiting the predictive power of individual neural network checkpoints to improve performance. Though the scope of this paper is limited to neural networks, **MoCE** can be used with any iteratively trained algorithm.

Materials and methods

In this section, we detail a study conducted with 14 patients experiencing epileptic seizures. Patients were equipped with wrist-worn wearable devices and monitored in clinic. Data from this study were used to examine the efficacy of our proposal.

Participants

Participants were recruited from Mayo Clinic, Rochester, and included patients undergoing video invasive EEG monitoring with stereoelectroencephalographic (sEEG) or subdural electrodes, as well as those admitted to the epilepsy monitoring unit for video scalp EEG monitoring with a high suspicion of seizures.

Ethics

This observational study was approved by the Mayo Clinic's Institutional Review Board after a full board review under protocol number 18-008357. All participants provided written informed consent prior to taking part in the study. All methods were performed in accordance with the relevant guidelines and regulations

Data collection

Data collection was conducted using research-grade, wrist-worn Empatica E4 devices while patients were simultaneously monitored via vEEG for an average of two to five days with continuous monitoring. The device records multiple physiological signals, including accelerometry (ACC), Blood Volume Pulse (BVP, derived from photoplethysmography), electrodermal activity (EDA), and skin temperature, while estimating Heart Rate (HR) from BVP. The sampling frequencies for these signals were 32 Hz for ACC, 64 Hz for BVP, 4 Hz for both EDA and temperature, and 1 Hz for HR⁹. The device is capable of recording for 36+ hours on a single charge⁴¹. In our study, features were limited to electrodermal activity and the accelerometry magnitude because prior work has shown these to be effective in detecting GTC seizures⁸.

The E4 devices were worn on the wrist, with placement prioritized to the side most involved during seizures when possible, as informed by prior clinical reports. When logistical constraints such as IV tubes interfered, the device was repositioned to avoid disruption to clinical care. Throughout participant's monitoring periods, a clinical coordinator was present to assist with any potential device issues.

Seizure onset and offset times were determined by institutional neurologists using video intracranial electroencephalography (iEEG) or scalp EEG monitoring. These expert annotations were used to label ictal segments (seizure segments) based on the corresponding timestamps. Data preprocessing and analysis were performed in Python (v3.11.5).

Data processing

Preprocessing of raw physiological signals acquired from Empatica E4 devices was performed to enable subsequent data partitioning. All data streams were converted to EDF format to recalibrate sampling frequencies and correct for timestamp drift, reducing errors from several minutes of drift to just a few seconds per day. To ensure consistency across data streams, all signals were resampled to a uniform rate of 32 Hz. Upsampling was achieved through linear interpolation, while downsampling utilized averaging. Non-wear periods were identified and excluded based on temperature readings, with values outside the 27–45 °C range classified as non-wear⁴². Additionally, data from the thirty-minute period following a seizure were removed to allow physiological parameters to stabilize and return to baseline. All signals were normalized within each patient by subtracting the mean and dividing by the standard deviation.

Data partitioning was performed by segmenting signals into 60-sec windows, resulting in segments with 1920 timesteps at the uniform sampling rate. Interictal segments (non-seizure segments) were partitioned into non-overlapping windows, while ictal segments were generated using a sliding window approach with a step size of 10 sec. This overlap strategy increased the number of ictal segments available for analysis, addressing class imbalance and capturing critical predictive features within individual windows. After partitioning, the dataset contained 52 ictal segments, derived from 23 seizures, and 90,967 interictal segments. For model training, interictal segments were undersampled to balance class distributions within each patient. In contrast, validation retained all interictal segments to provide a comprehensive evaluation of model performance.

Mixture of checkpoint experts

MoCE is a unique architecture in which a single expert is selected from a pool of experts to make the final prediction. MoCE differs from traditional Mixture of Experts approaches in that, instead of fitting multiple models, the expert pool consists of checkpoints from a single-pass neural network. Checkpoints are selected to ensure diversity while retaining performance. By selecting a single checkpoint, MoCE is able to introduce some interpretability into an otherwise black-box approach.

MoCE consists of 3 main components: checkpoint experts, assignment meta-learners, and the expert filter. A high-level workflow illustrating how these components connect in MoCE is provided in Fig. 1. The expert filter is applied on the meta-learners during model training to ensure an efficient and diverse set of experts. The meta-learners are then used at inference to assign predictions to individual experts. The full training procedure is detailed in Algorithm 1.

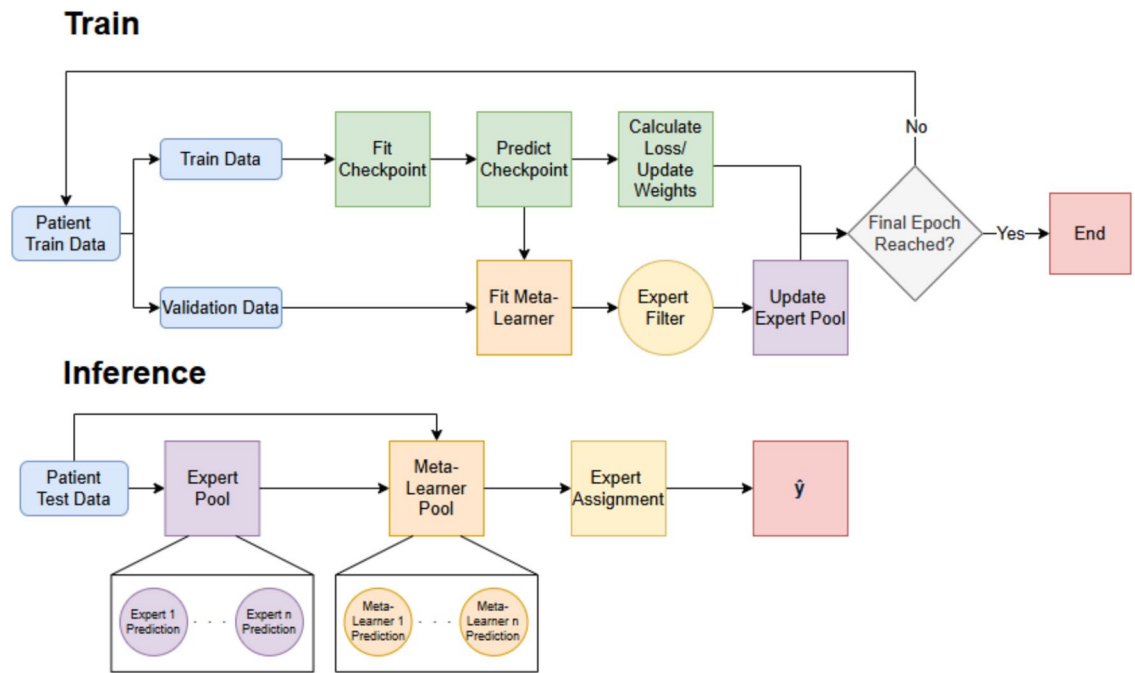


Fig. 1. Overall workflow of Mixture of Checkpoint Experts (MoCE). During training, the train data is split into a train and validation set. The train set is used to fit the checkpoints and calculate the loss for backpropagation. The validation set is used to fit the meta-learners. The meta-learners are passed to the expert filter which determines whether or not the checkpoint should be added to the expert pool. This process repeats once per epoch. During inference, the experts and meta-learners each make a prediction for every sample. These predictions are used to assign an individual expert per sample.

Require: n : Number Epochs, t : Similarity Threshold, X , y
 1: $E = []$; $M = []$;
 2: **for each** n **do**
 3: Split X into X_{train} , $X_{validation}$
 4: Predict X_{train}
 5: Calculate loss
 6: Backpropagate and update weights
 7: $\hat{y} = \text{Predict } X_{validation}$
 8: $m_i = \text{Train Meta Learner } (X_{validation}, \hat{y}, y, t)$
 9: $E, M = \text{Expert Filter}(m_i, E, M)$
 10: **end for**

Algorithm 1. Mixture of Checkpoint Experts (MoCE)

Checkpoint Experts. MoCE utilizes an interpretable Mixture of Experts framework. Multiple experts are fit with the training data and a single expert is chosen as the final predictor for each individual sample in the test set. In MoCE, instead of independently training multiple models, the experts are made up of a collection of various checkpoints in a single neural network. In this context, we define “checkpoint” as the frozen state of a model including all its neurons and their respective weights at a given epoch. In essence, each sample in the test data is predicted by a different stage of the neural network training process, attempting to identify the patterns best-suited for an individual sample.

Meta-Learners. A logistic regression is implemented as a meta-learner for each expert. In our case, the meta-learners are used for error anticipation. They identify the best expert for a given sample by predicting the likelihood that the expert will make a correct prediction. Samples are assigned to experts based on the meta-learner prediction. Inputs to the meta-learners include the feature set used to train the initial model and the expert’s predicted value for each training observation. The model’s outcome variable is a binary value indicating whether the expert correctly classified the observation, resulting in the meta-learner’s output being the probability that the expert will predict a given sample correctly. The addition of these transparent models to the framework, provide insights such as feature importance and directionality, offering necessary context towards understanding the model’s behavior (see Section “Results”).

Expert Filter. Throughout the model training process, model weights may not vary significantly between each training epoch, potentially resulting in an abundance of highly similar, redundant checkpoints, or experts. To avoid this inefficiency, we apply an expert filter to our checkpoints, determining whether each checkpoint should be saved or discarded, reducing the total number of checkpoints to a small, diverse, pool of experts.

The filtering algorithm proceeds as follows and an overview is provided in Algorithm 2. When a new meta-learner is fit, we take the Pearson Correlation between its coefficients and the coefficients of all meta-learners currently in our pool of experts. If the maximum correlation with any of the experts is below a user-specified threshold then the checkpoint is added to the pool of experts. If not, then the checkpoint's performance is compared against the performance of the existing expert with the highest correlation and the better performing checkpoint is kept while the other is discarded. The intuition is that the meta-learners provide us insight into what features the experts are using to make their predictions and similar coefficients would indicate similar behavior. The objective is to diversify the experts while improving predictive performance.

Require: M : Meta-Learner Pool, E : Expert Pool, m_i : Meta-Learner Model, e_i : Checkpoint at epoch i , t : Similarity Threshold

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1:  $r = \max_{m_j \in M} \left( \frac{\text{cov}(m_i, m_j)}{\sigma_{m_i} \sigma_{m_j}} \right)$  ▷ Pearson Correlation of Model Coefficients
2: if  $r < t$  then
3:    $E.\text{append}(e_i)$ 
4:    $M.\text{append}(m_i)$ 
5: else
6:    $k = \text{argmax}_{m_j \in M} \left( \frac{\text{cov}(m_i, m_j)}{\sigma_{m_i} \sigma_{m_j}} \right)$ 
7:   if  $\text{loss}(e_i) < \text{loss}(e_k)$  then
8:      $E[k] = e_i$ 
9:      $M[k] = m_i$ 
10:  end if
11: end if
12: return  $E, M$ 

```

Algorithm 2. Expert Filter

Prediction Assignment. At inference time, the meta-learners predict the likelihood that the underlying expert will make a correct prediction for an individual sample. The sample is then assigned to the expert with the highest predicted probability. Formally, this can be represented as

$$e = \underset{m_i \in M}{\text{argmax}} [m_i(x)] \quad (1)$$

where x is a sample in the test dataset and M is the collection of meta-learners.

The model then returns the output of the selected expert e from the pool of experts E as the final prediction:

$$\hat{y} = E_e(x) \quad (2)$$

Intuitively, each sample is being predicted by a single expert. The meta-learners guide the model towards the best expert for that sample.

Experimental design

We compare the performance of **MoCE** using both an LSTM and CNN to the baseline neural networks. We chose LSTM and CNN as these are foundational Deep Learning (DL) models in time-series classification, both of which have been widely applied to seizure detection and other wearable device applications in the past. Importantly, these experiments can be extended onto any iteratively trained DL algorithm. Additionally, we compared our model against four traditional ML algorithms commonly used for detecting seizures using EEG recordings, specifically Logistic Regression, Decision Tree, SVM, and KNN. The models were tested using leave-one-out cross validation across 14 patients. The experiments were run 10 times for a total of 140 trials. In each trial, data was split into a train and validation set and 3-fold cross validation was used to select hyper-parameters. The parameters searched are included in Table 2.

We measure performance by focusing on the model's ability to identify seizures while avoiding false positives (FP). Specifically, we use recall, area under the receiver operating characteristic curve (AUC), seizure-level recall, and False Alarms per Hour (FAPH). Recall measures the model's ability to correctly identify each unique time step as ictal or interictal. Seizure-level recall, on the other hand, takes the collection of unique timesteps belonging to a single seizure, and collapses them into a single true positive (TP) or false negative (FN) depending on whether the model successfully identified *any* of the timesteps that occurred during the seizure as an ictal segment. This metric provides a sense of how well the model correctly identifies when a seizure is occurring instead of the proportion of seizure segments correctly identified. FAPH is the number of times per hour the model would, in theory, incorrectly report a seizure, based on the false positive rate.

Next, we conduct an ablation study comparing three alternative versions of **MoCE**. In the first proposal, we remove the meta-learners and prediction assignment and treat the experts as an ensemble that predicts the most common prediction from all the experts (**MoCE w/o ML**). Additionally, we examine the effect of the expert filter by considering the performance of **MoCE** in which we save the checkpoint at every epoch for our expert pool

Number of participants	Total N (%)
	14 (100)
Sociodemographic variables	
Age (years), median (IQR)	31 (18.75, 38.50)
Sex, N (%)	
Female	5 (36)
Male	9 (64)
Race, N (%)	
American Indian/Alaskan Native	1 (7)
White	12 (86)
Not reported	1 (7)
Ethnicity, N (%)	
Not Hispanic or Latino	14 (100)
Clinical variables	
Number of seizures (IQR)	1.5 (1, 2)
Seizure duration, seconds (IQR)	130 (102, 154)

Table 1. Overview of study cohort demographics.

Model	Parameter	Values
LSTM	Hidden Size	[32, 64]
	Number Layers	[2, 3, 4]
	Dropout Probability	0.2
	Number Epochs	[25, 100, 200]
	Learning Rate	[0.001, 0.01]
CNN	Number Filters	32
	Kernel Sizes	[4, 8]
	Pooling Sizes	[2, 4]
	Dropout Rate	0.2
	Learning Rate	[0.001, 0.01]
	Number Epochs	[25, 100, 200]
MoCE	Similarity Threshold	[0.1, 0.2, 0.3, 0.4, 0.5]

Table 2. Overview of the tuning parameters searched during model training.

(MoCE w/o EF). Finally, we isolate the impact of the base model's prediction on the meta-learner's performance. In this experiment, we retrained the entire MoCE model except the meta-learners were trained on the original feature set excluding the checkpoint's predicted value (MoCE w/o Pred.).

Participant Population The study included 14 adults with GTC seizures. The median participant age was 31 (IQR, [18.75, 38.50]) and 36% were women (see Table 1). On average, each patient experienced 1.5 seizures (IQR, [1, 2]), with each seizure lasting on average 130 seconds (IQR, [102, 154])

Results

In this section we report the findings of our experimental evaluation and use a case study to explore the benefits of the MoCE's explanations for both data scientists and clinicians.

Overall performance

RQ1: "Does MoCE improve the predictive performance of a standard black-box neural network?"

The results of our experimental evaluation are displayed in Table 3. Consistent with prior work, the neural network architectures are superior to traditional ML methods in detecting seizures. For example, both the baseline LSTM and CNN have recalls nearly double that of Logistic Regression. The best performing traditional model is SVM. However, in addition to not being interpretable, SVM's average recall is 5% worse than LSTM and 9% worse than CNN.

MoCE has better false positive rates than the baseline neural networks reducing the number of false alarms per hour by an average of 14.5%. Meanwhile, performance in recall and AUC is within the margin of error. These findings suggest that our model is able to identify seizures as good as the baseline approach with fewer false positives, an overall improvement in performance.

We further analyze these results using a Bayesian sign test to demonstrate the significance of our findings^{43,44}. The results of this test are presented in Fig. 2. Again these findings illustrate that our approach virtually ties the

	Traditional ML models				
	Decision tree	Logistic regression	KNN	SVM	
Recall (↑)	0.516 ± 0.095	0.380 ± 0.063	0.117 ± 0.050	0.699 ± 0.096	
AUC (↑)	0.612 ± 0.046	0.484 ± 0.067	0.656 ± 0.051	0.763 ± 0.074	
Seizure Recall (↑)	0.690 ± 0.112	0.595 ± 0.109	0.232 ± 0.100	0.857 ± 0.082	
FAPh (↓)	19.334 ± 1.877	13.666 ± 1.345	2.811 ± 0.499	14.980 ± 1.630	
	LSTM				
	Baseline	MoCE	MoCE w/o ML	MoCE w/o EF	MoCE w/o Pred.
Recall (↑)	0.734 ± 0.027	0.731 ± 0.025	0.783 ± 0.026	0.713 ± 0.027	0.742 ± 0.025
AUC (↑)	0.795 ± 0.016	0.806 ± 0.014	0.734 ± 0.014	0.771 ± 0.015	0.805 ± 0.014
Seizure Recall (↑)	0.890 ± 0.023	0.878 ± 0.023	0.928 ± 0.020	0.873 ± 0.023	0.889 ± 0.022
FAPh (↓)	19.778 ± 1.008	15.871 ± 0.566	18.820 ± 0.613	14.327 ± 0.462	16.514 ± 0.574
	CNN				
	Baseline	MoCE	MoCE w/o ML	MoCE w/o EF	MoCE w/o Pred.
Recall (↑)	0.768 ± 0.027	0.804 ± 0.022	0.821 ± 0.023	0.764 ± 0.026	0.796 ± 0.023
AUC (↑)	0.752 ± 0.014	0.796 ± 0.013	0.747 ± 0.011	0.774 ± 0.016	0.792 ± 0.014
Seizure Recall (↑)	0.905 ± 0.023	0.951 ± 0.015	0.961 ± 0.015	0.918 ± 0.020	0.954 ± 0.015
FAPh (↓)	19.685 ± 1.051	17.830 ± 0.638	19.825 ± 0.890	17.428 ± 0.537	17.656 ± 0.635

Table 3. Comparison between traditional ML methods, baseline neural networks, and MoCE. Results include our ablation study. MoCE w/o EF refers to MoCE with the expert filter removed and MoCE w/o ML is MoCE with the meta-learners removed. MoCE w/o Pred. uses the entire MoCE pipeline except the meta-learners are trained without knowledge of the checkpoint's prediction. Average score and standard error of MoCE across all 140 trials in Recall, Area Under the Receiver Operating Characteristic Curve (AUC), Seizure-level Recall, and False Alarms per Hour (FAPh). $\uparrow$ indicates a higher score is better while $\downarrow$ indicates a lower score is better. The best score is indicated in bold.

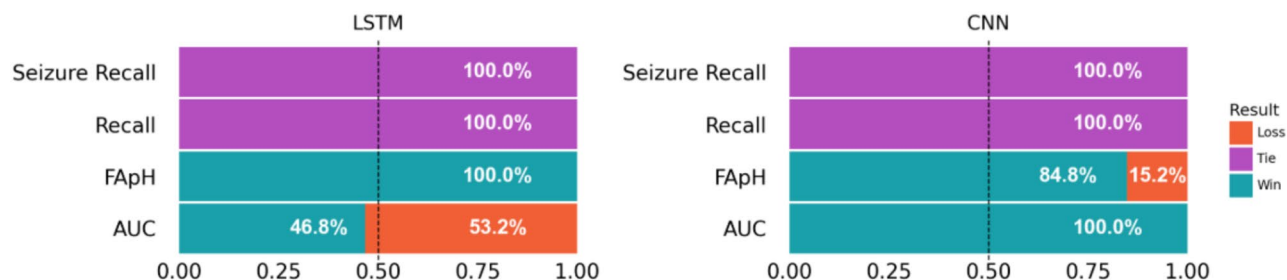


Fig. 2. Results of the Bayesian Sign Test comparing MoCE against the classical neural network. Win indicates MoCE performed better than the neural network. Ties are defined using a practical equivalence interval of $[-1\%, 1\%]$.

baseline in recall while significantly improving the false alarm rate. Additionally, our approach is competitive with the standard neural network architecture in AUC. These results suggest that it is more probable than not that our proposal's performance is a meaningful improvement over the baseline neural networks.

Next, we consider the results of our ablation study shown in Table 3. The full **MoCE** proposal outperforms the alternatives in AUC. MoCE w/o ML is the best in terms of recall and seizure-level recall but at the expense of a much higher FAPh. Removing the baseline model's predicted value from the meta-learner features resulted in minimal changes across all relevant metrics. These results suggest there is no meaningful difference between these two models and that multiple meta-learning approaches may be viable. Concerning RQ1, we find that **MoCE** is able to match or improve upon the performance of a typical black-box neural network.

Model explanation case study

In addition to improvements in predictive performance, **MoCE** is beneficial because it provides insights into the behavior of the model. **MoCE** allows for both global explanations, looking for overall patterns in the performance of the model, and local explanations, providing details on the rationale for a specific prediction. We present a case study demonstrating how **MoCE**'s explanatory power could be used in a real-world setting. For this case study we examine the predictions on the patient with the most seizures in the dataset.

RQ2: "Can **MoCE** be used to understand a neural network's global decision-making patterns?"

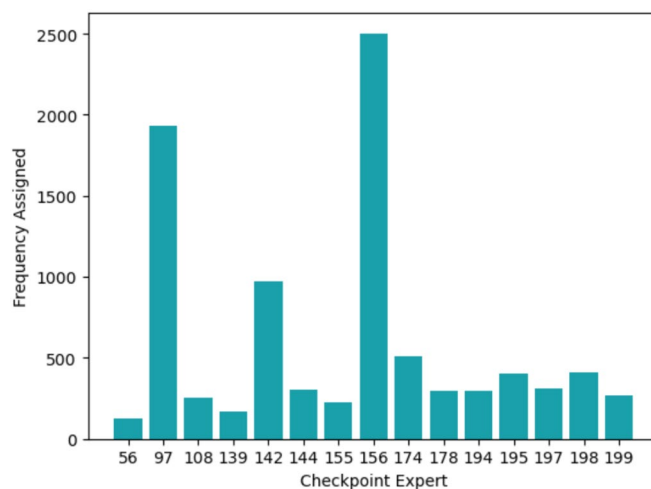
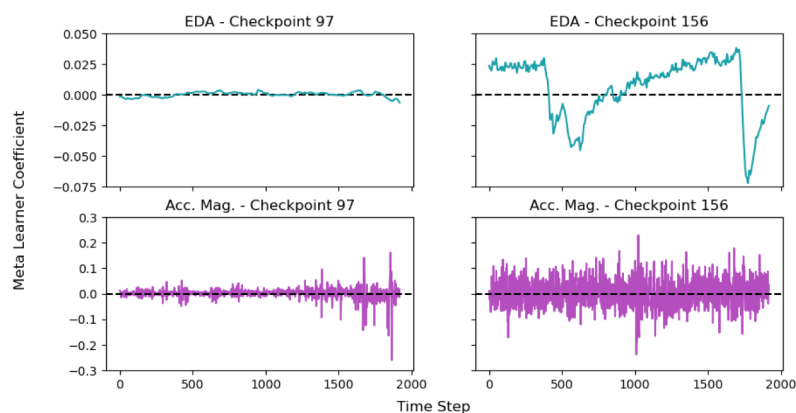


Fig. 3. Frequency that each expert was assigned to make the final prediction.

(a) β coefficients of the logistic regression meta-learner. The x-axis corresponds to the timesteps in the data. Features are separated into electrodermal activity (EDA) and Accelerometry Magnitude (Acc. Mag.)



(b) Actual patient data in Accelerometry Magnitude (Acc. Mag.) and electrodermal activity (EDA) for the least confident True Positive (TP) and False Positive (FP) case. The x-axis corresponds to the timesteps in the data.

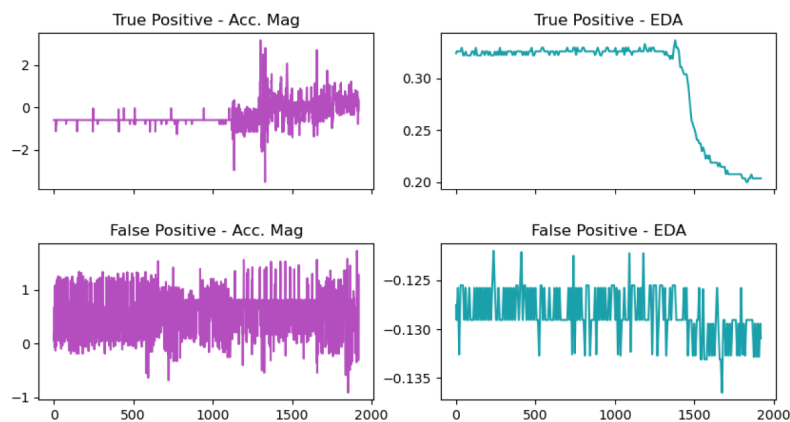


Fig. 4. Comparison of patient data to logistic regression meta-learners.

	Meta-learner	Model	Acc. Mag.	EDA
Result	Prediction	Prediction	Weight	Weight
TP	1.000	0.538	4.160	1.536
TP	0.998	0.587	0.926	0.624
TP	0.999	0.852	− 2.348	4.617
TP	0.999	0.871	0.629	− 0.065
TP	0.999	0.935	0.674	9.707
FP	1.000	0.500	5.922	− 1.407
FP	1.000	0.500	− 1.113	2.680
FP	0.999	0.501	1.330	0.365
FP	0.999	0.502	1.695	0.216
FP	0.998	0.503	0.818	− 0.033

Table 4. Meta-Learner predictions and feature weights for the 5 True Positive (TP) and False Positive (FP) predictions nearest to the decision boundary. Weights greater than 0 indicate that the features increased the likelihood of a correct prediction according to the meta-learner.

Global Explanations. Figure 3 illustrates how frequently each expert is selected. The model is relying on two of the 15 experts (checkpoints 97 and 156) for a majority of the predictions (~62.8%). Focusing on these two experts, we can begin to search for global patterns in the model's behavior.

Figure 4a shows how the coefficients on EDA and accelerometry magnitude change over time for these two checkpoints. Using these we are able to identify which patterns in the data an individual checkpoint specializes. The more closely the data match the coefficients, the more likely the checkpoint is to correctly predict the sample. A positive coefficient implies that the model is more likely to be correct for positive values in the underlying feature.

These coefficients provide insights into the rationale for the model's prediction that are unavailable in the standard black-box architecture. For example, in Checkpoint 97, the coefficients for EDA are consistently at or around zero suggesting that the model is not relying heavily on these features. Meanwhile, Checkpoint 156 is more likely to be correct when the EDA has a higher variance but within a consistent pattern of starting high and then slowly increasing. In the black-box architecture, we are unable to directly identify how individual features contribute towards the final prediction. Instead, it would be necessary to fit a post-hoc explanation model which is necessarily biased and may not accurately represent the model's process. With regards to RQ2, we find that **MoCE** allows for a data scientist to observe overall patterns within the model that would otherwise not be possible with a classical neural network.

RQ3: "Is **MoCE** able to explain the rationale for individual predictions in a way that is beneficial to both data scientists and clinicians?"

Local Explanations. Using the learned coefficients from the meta-learners, we can analyze individual predictions to determine how much importance the selected expert is placing on each feature for individual patients. We separate the coefficients based on feature type (Acc. Mag./EDA), and sum the impact each is having towards the meta learner's prediction. Positive values suggest that the checkpoint is more likely to make an accurate prediction based on that feature.

Table 4 contains the weight applied to the accelerometry magnitude and EDA by the assigned expert for the 5 true positive and false positive cases with the lowest predicted probabilities. In each of these cases, the meta-learner is confident the underlying expert is correct in its prediction, but for different reasons. In 3 of the 5 TP cases, the meta-learner believes the checkpoint is probably correct both because of the EDA and accelerometry (weights greater than 0). In the other cases, the meta-learner believes the model is well-equipped to handle either the accelerometry or the EDA for that particular case but not both.

Examining these cases further, Fig. 4b shows the patient data for the TP and FP case with the lowest predicted probability. The accelerometry has a large magnitude and high variance in both cases which are signals consistent with symptoms of a GTC seizure. On the other hand, EDA in the TP case starts high and remains static over most of the observation period before dropping off, consistent with behavior from a GTC seizure. By contrast, the EDA in the FP case has a much smaller, negative magnitude. The meta-learners indicate high confidence that both checkpoints are predicting cases with this accelerometry correctly and that the TP expert is handling the EDA correctly. On the other hand, the meta-learners indicate low confidence that the FP expert is handling the EDA data correctly, consistent with our observations.

Pairing the coefficients with the data, we can not only understand why the model is making an individual prediction but also begin to recognize patterns that may be suggestive of false positives or false negatives. Within the traditional black-box architecture, this analysis could only be performed on the data without the direct model insights potentially leading to biased analysis. Concerning RQ3, these results demonstrate that **MoCE** allows for detailed explanations behind the rationale for individual predictions.

Discussion

As the results indicate, **MoCE** can be combined with a neural network to improve its underlying performance. Neural networks specialize in identifying complex patterns in the data, but as the model evolves over time, some

of these patterns are forgotten even though they may be important for individual predictions. By introducing checkpoint experts, we are able to retain these patterns and identify niche cases that still require them. In our case study, **MoCE** relies heavily on a few experts for the majority of cases, but retains specialized experts for classifying niche cases. By adding **MoCE** to a neural network, we can detect more seizures while decreasing the number of false positives.

In addition, using neural networks for high stakes decision-making such as healthcare can be problematic because their lack of transparency does not allow for auditing mistakes. While explanation techniques have been proposed in the past, these tend to suffer from their own biases and can misrepresent the model. It is not possible for explanation techniques to be fully faithful to the underlying black-box model otherwise the explanation model could replace the black-box²⁵. In contrast, **MoCE** provides direct insight into the model's behavior. By using fully interpretable Logistic Regression meta-learners, **MoCE** is able to fully trace the rationale behind an assigned expert and identify which features an expert is good or bad at predicting. With this added transparency, **MoCE** provides extra information that is useful for both the clinician and the data scientist.

Data Scientist Workflow Using global explanations, a data scientist can assess the performance of specific experts and evaluate their strengths and weaknesses. Using this information, a data scientist can then fine tune the model to improve its predictive power. For example, if a specific checkpoint is generating too many false positives in situations when a patient's accelerometry exhibits high variance but low magnitude, the data scientist could find specific counterexamples in the training data and further train the checkpoint to better discriminate true seizures. Alternatively, if the data scientist observes that one poor performing checkpoint is being assigned too many cases, they could attempt to adjust the meta-learner output to better reflect performance or, if necessary, even discard the checkpoint entirely. With the global explanations, data scientists are better able to pinpoint model weaknesses and know where to focus their effort on improving its performance.

Clinician Workflow Further, using the local explanations, data scientists have more information to flag specific high-risk cases for a clinician to examine. The local explanations suggest why the model thinks a specific prediction is correct. In the case study, the meta-learners capably identified that the EDA for the two cases was a discriminating factor even though the underlying model was uncertain in its prediction. With this information, the clinician has an understanding of not only when a seizure was detected but why. These features may be informative clinically; for example, EDA and HR changes may indicate autonomic seizure involvement, or a sudden, rapid increase in apparent seizures might actually be an artifact of a separate clinical change or medication adjustment. The model's explanations may help to guide the search for effective medication, neuromodulatory, or surgical treatments.

Quality interpretations should be model-specific and there is no single solution⁴⁵, but with the meta-learners, both the data scientist and clinician have more details to inform their decision beyond the model's raw output.

Limitations The results in this paper provide evidence of a positive outlook on the generalizability of our proposal. Our proposal has half the variance of the baseline models in terms of FApH, and statistical testing suggests a high level of confidence in our proposal's performance relative to the baseline. That being said, we acknowledge that the sample size of our dataset is limited, including only 14 patients and 23 observed seizures during in-clinic stays. As a result, the feasibility of **MoCE** as a seizure detection method should be validated against large, heterogeneous populations before considering integration into clinical workflows. Additionally, while the logistic regression meta-learners are sufficient for improving upon the baseline neural networks, the expert assignment can be improved in our model. In practice, a data scientist could use the information gathered from the model performance over time to refine the expert assignment process while retaining interpretability. Future work will focus on techniques to improve meta-learner performance.

Conclusion

In this paper we propose **MoCE**, Mixture of Checkpoint Experts, a novel machine learning technique designed to introduce partial transparency to any neural network framework. Through a robust experimental evaluation, we demonstrate that our approach is capable of using wrist-worn wearable devices to detect seizures better than existing neural networks. More importantly, **MoCE** offers insights into a model's behavior that are beneficial to both a data scientist and medical professional in a clinical setting. With **MoCE**, data scientists can more easily evaluate decision-making patterns and use these to further refine model performance. Furthermore, data scientists and clinicians are able to audit individual predictions and diagnose why a prediction was made providing greater information about the model and its behaviors. **MoCE** directly addresses the issues of black-box models that many clinicians face while improving upon their predictive performance.

Data availability

The datasets generated and/or analysed during the current study are not publicly available due to it containing participant's protected health information but are available from the corresponding author on reasonable request.

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Author contributions

JG developed and designed the algorithm. BB, JC, LF, and EF oversaw the project. BB, NC, JC, LF, EF, and NM provided significant mentoring and feedback throughout the project. JG and LF managed the writing of this manuscript. All authors reviewed and approved the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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